

Regular Labor: Chilean Prune Shocks, Digestive Health, and Earnings

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Abstract

This paper studies how short-run disruptions to digestive health affect labor-market outcomes. We use weather shocks in Chilean prune-producing regions as an instrument for Danish prune supply, exploiting Denmark’s reliance on Chile as its largest direct source of dried prunes. A one-standard-deviation adverse weather shock in Chile reduces Danish prune imports by 4.6 percent and raises retail prune prices by 2.5 percent. Under the exclusion restriction that Chilean growing-season weather affects Danish earnings only through prune availability, the IV estimates imply that a 10 percent increase in prune prices raises constipation-related primary-care contacts by 3.2 percent and reduces monthly earnings by 0.12 percent. The earnings effects operate through small increases in sickness absence and reductions in paid hours, and are concentrated among older workers, workers with prior constipation-related health contacts, and occupations with limited schedule flexibility. Results are similar, though less precise, when we instrument for Danish prune availability using weather shocks in California’s Central Valley, the largest global prune-producing region. The findings suggest that a strategic prune reserve would be a low-cost, if unusually specific, form of labor-market stabilization policy.

Keywords: digestive health, labor supply, earnings, weather shocks, instrumental variables, prunes

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*All estimates in this draft are illustrative. Strategic prune reserve policy simulations are, at present, aspirational.

†Correspondence welcome, especially from readers with strong priors on the exclusion restriction.